EEC 201
Computer Assignment 3

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1 Sixth-Order Butterworth Lowpass IIR Filter

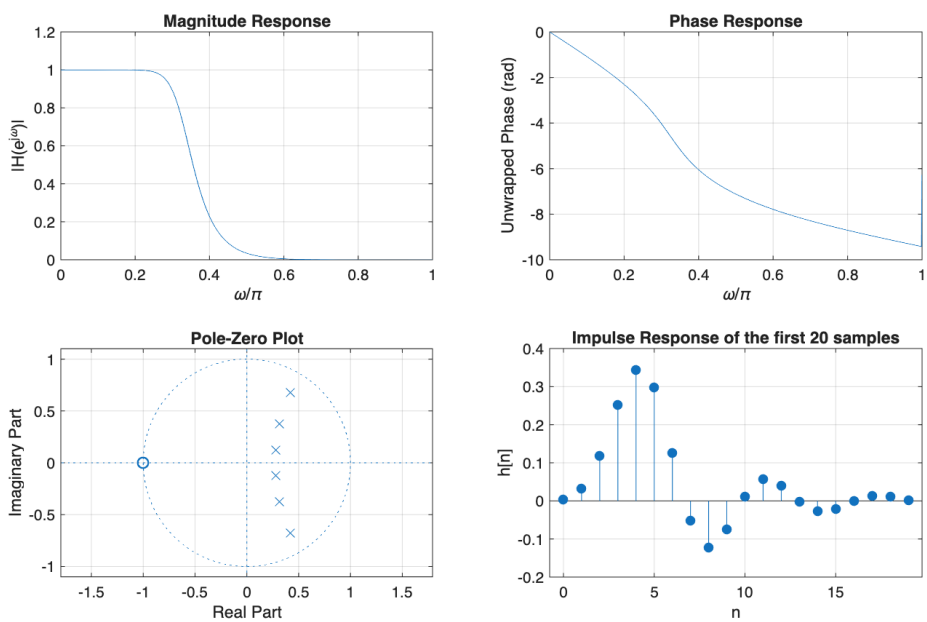


Figure 1: Plots generated

First 20 samples of h[n]:

n	h
0	0.0041063
1	0.032938
2	0.11868
3	0.2521
4	0.34429
5	0.29828
6	0.12534
7	-0.052697
8	-0.12217
9	-0.074247
10	0.011878
11	0.056055
12	0.039423
13	-0.0023729
14	-0.026835
15	-0.020816
16	-0.00031071
17	0.012945
18	0.010966
19	0.00090405

Figure 2: First 20 values of the impulse response

Code

A 6th-order lowpass Butterworth filter was designed using butter(N, Wn, 'low'), where N = 6 and Wn = 0.33 (normalized so 1 corresponds to π rad/sample) as written on the assignment. This returned the numerator and denominator coefficients b and a of the digital IIR filter.

I calculated the frequency response with freqz(b,a) to get the magnitude and phase. The pole-zero locations were found via roots(b) and roots(a) and visualized with zplane, and the first 20 impulse response samples were generated using impz(b,a,20) and plotted with stem.

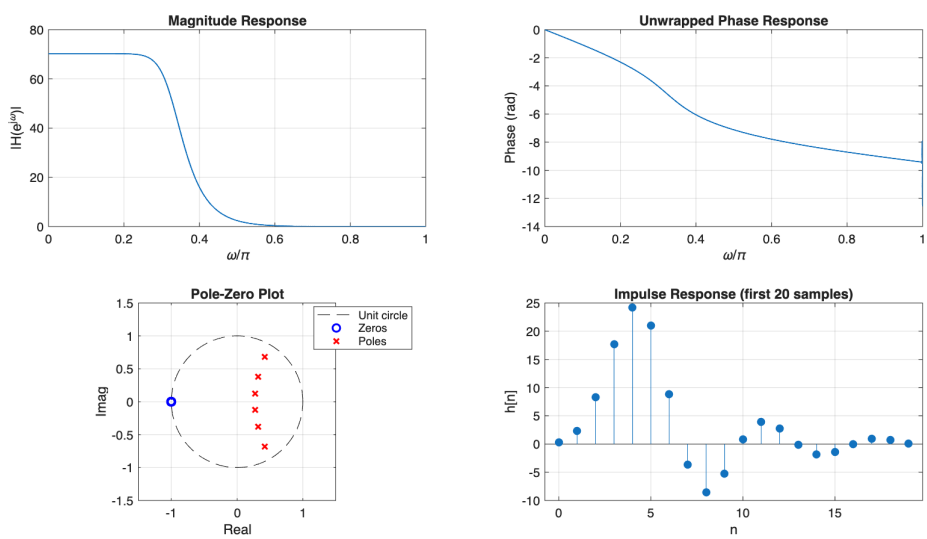
```
%% Question 1: Analyzing a Sixth-Order Digital Butterworth Lowpass Filter
% using 'butter' to design a 6th-order LP IIR filter with wc = 0.33pi
% Wn = 0.33

clear; close all; clc;

N  = 6;
```

```
Wn = 0.33;
[b,a] = butter(N, Wn, 'low');


%% (a) Calculating the frequency response
% magnitude + phase

nfft = 2048;
[H,w] = freqz(b,a,nfft);

mag = abs(H);
mag_db = 20*log10(max(mag,1e-12));
ph  = unwrap(angle(H));

%% (b) Pole-zero plot
% b = numerator
% a = denominator

z = roots(b);
p = roots(a);

%% (c) Impulse response (first 20 values)
% 0 ~ 19 = 20

n  = 0:19;
h  = impz(b,a,20);


%% (d) Plotting Everything

figure('Name','6th-Order Butterworth Lowpass IIR','Color','w');

subplot(2,2,1);
plot(w/pi, mag); grid on;
xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
title('Magnitude Response');

subplot(2,2,2);
plot(w/pi, ph); grid on;
xlabel('\omega/\pi'); ylabel('Unwrapped Phase (rad)');
title('Phase Response');

subplot(2,2,3);
zplane(z,p); grid on;
title('Pole-Zero Plot');
```

```
subplot(2,2,4);
stem(n, h, 'filled'); grid on;
xlabel('n'); ylabel('h[n]');
title('Impulse Response of the first 20 samples');

%% Printing the first 20 impulse samples
disp('First 20 samples of h[n]:');
disp(table(n(:), h(:), 'VariableNames', {'n','h'}));
```

Extension

Although I didn't have enough time to try this with the rest of the filters, I wanted to use your feedback from class and try to fully understand the Butterworth filter by eliminating the usage of the Signal Processing Tool Box.
The output was as follows:



Code

I first specified the 6th-order and normalized cutoff, then I prewarped the digital cutoff using the bilinear transform formula so I could design the equivalent analog Butterworth filter. I found the analog poles on a circle in the left-half plane, mapped them into the z-plane using the bilinear transform, placed zeros at $z = -1$, and solved for the gain to make the DC response equal to 1 before forming the numerator and denominator polynomials.

After that, I tried to find the frequency response directly from the coefficients using the definition of $H(e^{j\omega})$.

```
%% Using no butter, freqz, zplane, impz
%  Spec from (1): N = 6, wc = 0.33*pi  -> Wn = 0.33 (normalized)

clear; close all; clc;

%% Filter parameters from the prompt
N  = 6;
Wn = 0.33;

wc = Wn*pi;

%% Prewarping the cutoff for the bilinear transform
% I used T = 1, so the bilinear prewarp is:  Wc = 2*tan(wc/2).

Wc = 2*tan(wc/2);               % Analog cutoff that maps to wc after bilinear transform.

%% Analog Butterworth poles
% I placed the N Butterworth poles uniformly on a circle of radius Wc,
% and I used the standard angle formula that gives only left-half-plane poles.

p = zeros(N,1);
for k = 0:N-1
    theta = pi*(2*k + 1 + N)/(2*N);
    p(k+1) = Wc * exp(1j*theta);
end

%% Analog poles into digital poles (bilinear) ---
% I used:  z = (2 + s)/(2 - s)
pd = (2 + p)./(2 - p);

%% Digital zeros
% A Butterworth lowpass has no finite analog zeros, so after bilinear transform
% the "zeros at infinity" map to zeros at z = -1 (repeated N times).

zd = -ones(N,1);

%% DC gain = 1
% I enforced H(z=1) = 1 by solving:
% H(1) = k * prod(1 - z_i^-1) / prod(1 - p_i^-1)  = 1.

k = real( prod(1 - pd.^(-1)) / prod(1 - zd.^(-1)) );

%% Zeros/poles/gain into polynomial b,a
b = k * poly(zd);
a = poly(pd);
b = real(b); a = real(a);
```

```
%% Frequency response (with no freqz) ----------
nfft = 2048;
w = linspace(0, pi, nfft).';

H = H_of_ejw(b, a, w);
mag = abs(H);
ph  = unwrap(angle(H));

%% First 20 impulse samples (no impz)
L = 20;
h = impulse_response(b, a, L);
n = (0:L-1).';

%% Poles and zeros (for plotting)
z = roots(b);
p2 = roots(a);

%% Magnitude, phase, pole-zero, and impulse
figure('Name','Toolbox-free Butterworth LPF (N=6, Wn=0.33)','Color','w');

subplot(2,2,1);
plot(w/pi, mag, 'LineWidth', 1); grid on;
xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
title('Magnitude Response');

subplot(2,2,2);
plot(w/pi, ph, 'LineWidth', 1); grid on;
xlabel('\omega/\pi'); ylabel('Phase (rad)');
title('Unwrapped Phase Response');

subplot(2,2,3);
pz_plot(z, p2);
title('Pole-Zero Plot');

subplot(2,2,4);
stem(n, h, 'filled'); grid on;
xlabel('n'); ylabel('h[n]');
title('Impulse Response (first 20 samples)');

%% First 20 impulse samples
disp('First 20 samples of h[n] (n = 0..19):');
disp(table(n, h, 'VariableNames', {'n','h'}));

%% Local Functions
```

```
function H = H_of_ejw(b, a, w)
%   H(e^{jw}) = (sum_k b(k) e^{-jw(k-1)}) / (sum_k a(k) e^{-jw(k-1)})

    nb = length(b);
    na = length(a);

    Eb = exp(-1j*w*(0:nb-1));
    Ea = exp(-1j*w*(0:na-1));

    Num = Eb * b(:);
    Den = Ea * a(:);

    H = Num ./ Den;
end

function h = impulse_response(b, a, L)
%   y[n] = sum_{k=0}^{Nb-1} b[k] x[n-k] - sum_{k=1}^{Na-1} a[k] y[n-k]
% with x[0]=1 and x[n>0]=0.

    nb = length(b);
    na = length(a);

    x = zeros(L,1); x(1) = 1;
    y = zeros(L,1);

    for n = 1:L
        acc_num = 0;
        for k = 1:nb
            idx = n - (k-1);
            if idx >= 1
                acc_num = acc_num + b(k)*x(idx);
            end
        end

        acc_den = 0;
        for k = 2:na
            idx = n - (k-1);
            if idx >= 1
                acc_den = acc_den + a(k)*y(idx);
            end
        end

        y(n) = (acc_num - acc_den) / a(1);
    end

    h = y;
```

```
end

function pz_plot(z, p)

    t = linspace(0, 2*pi, 400);
    plot(cos(t), sin(t), 'k--'); hold on;
    grid on; axis equal;

    if ~isempty(z)
        plot(real(z), imag(z), 'bo', 'LineWidth', 1.5);
    end
    if ~isempty(p)
        plot(real(p), imag(p), 'rx', 'LineWidth', 1.5);
    end

    xlabel('Real'); ylabel('Imag');
    legend('Unit circle','Zeros','Poles','Location','best');
    xlim([-1.5 1.5]); ylim([-1.5 1.5]);
end
```

2 Sixth-Order Digital Chebyshev Lowpass IIR Filter

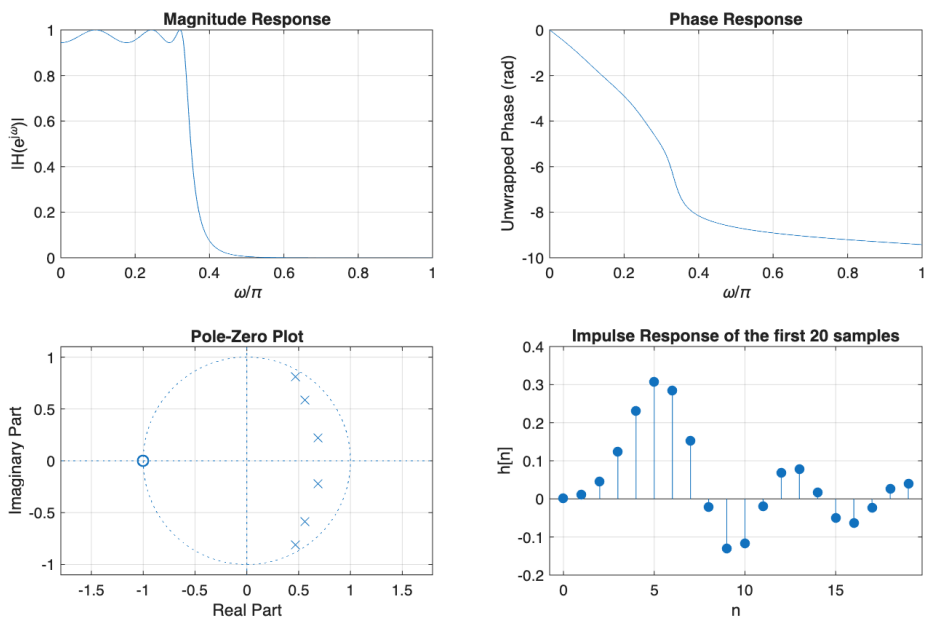


Figure 3: Plots generated

```
First 20 samples of h[n] (n=0..19):
    n         h
    __    ________

     0     0.001094
     1     0.010326
     2     0.045411
     3      0.12341
     4      0.23047
     5      0.30722
     6      0.28525
     7       0.1522
     8    -0.021319
     9      -0.1297
    10     -0.11674
    11    -0.019412
    12     0.068541
    13     0.078024
    14     0.016833
    15    -0.049517
    16     -0.06319
    17    -0.022535
    18     0.026366
    19      0.03893
```

Figure 4: First 20 values of the impulse response

Code

I used (as instructed), cheby1(N, Rp, Wn, 'low'), where N = 6, cutoff Wn = 0.33, and PB ripple was 0.5 dB. I found the filter coefficients b and a, and I then calculated the frequency response with freqz to get the magnitude and phase. Poles and zeros were once again found using roots(), and the first impulse response too was generated once again using impz(b,a,20).

```
%% Question 2: Analyzing a Sixth-Order Digital Chebyshev Lowpass IIR Filter
% command to use: cheby1
% wc = 0.33pi, PB ripple = 0.5 dB
% plotting mag + phase frequency responses, pole/zero, 20 vals if h

clear; close all; clc;

N  = 6;
Wn = 0.33;
Rp = 0.5;

[b,a] = cheby1(N, Rp, Wn, 'low');

%% Frequency response
```

```
nfft = 2048;
[H,w] = freqz(b,a,nfft);

mag    = abs(H);
mag_db = 20*log10(max(mag,1e-12));
ph     = unwrap(angle(H));

%% Pole-zero locations

z = roots(b);
p = roots(a);

%% Impulse response (first 20 samples)

n = 0:19;
h = impz(b,a,20);

%% Plots

figure('Name','6th-Order Chebyshev Type I Lowpass IIR','Color','w');

subplot(2,2,1);
plot(w/pi, mag); grid on;
xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
title('Magnitude Response');

subplot(2,2,2);
plot(w/pi, ph); grid on;
xlabel('\omega/\pi'); ylabel('Unwrapped Phase (rad)');
title('Phase Response');

subplot(2,2,3);
zplane(z,p); grid on;
title('Pole-Zero Plot');

subplot(2,2,4);
stem(n, h, 'filled'); grid on;
xlabel('n'); ylabel('h[n]');
title('Impulse Response of the first 20 samples');

disp('First 20 samples of h[n] (n=0..19):');
disp(table(n(:), h(:), 'VariableNames', {'n','h'}));
```

3 Sixth-Order Digital Elliptic Lowpass IIR Filter

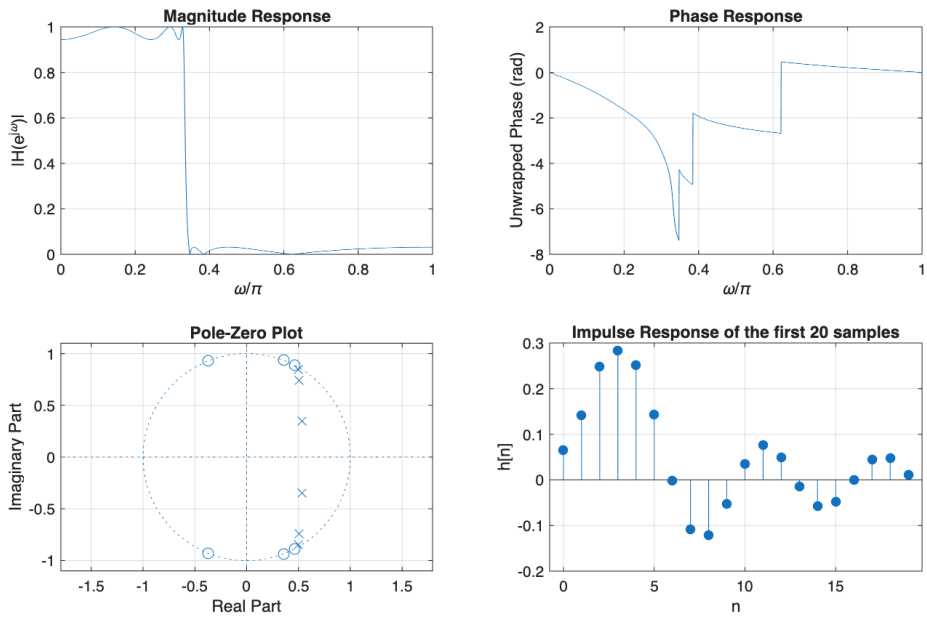


Figure 5: Plots generated

```
First 20 samples of h[n]:
    n          h
    __    __________

     0       0.065249
     1         0.1419
     2        0.24799
     3        0.28392
     4        0.25182
     5        0.14286
     6     -0.0021577
     7       -0.10781
     8       -0.12127
     9      -0.052678
    10       0.035258
    11       0.076558
    12       0.049394
    13      -0.013748
    14      -0.057391
    15      -0.048506
    16    -5.1763e-05
    17       0.043968
    18       0.047185
    19       0.010359
```

Figure 6: First 20 values of the impulse response

Code

The code was very similar to questions 1 and 2, except here, the stopband attenuation was Rs = 30 dB.

```
%% Question 3: Analyzing a Sixth-Order Digital Elliptic Lowpass IIR Filter
% command to use: ellip
% this time, SB ripple is 30 dB less than PB.
% so stopband attenuation 30 dB below passband level

clear; close all; clc;

N  = 6;
Wn = 0.33;
Rp = 0.5;
Rs = 30;

[b,a] = ellip(N, Rp, Rs, Wn, 'low');

%% Frequency response

nfft = 2048;
```

```
[H,w] = freqz(b,a,nfft);

mag = abs(H);
ph  = unwrap(angle(H));

%% Pole-zero locations

z = roots(b);
p = roots(a);

%% Impulse response (first 20 samples)

n = 0:19;
h = impz(b,a,20);

%% Plots

figure('Name','6th-Order Elliptic Lowpass IIR','Color','w');

subplot(2,2,1);
plot(w/pi, mag); grid on;
xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
title('Magnitude Response');

subplot(2,2,2);
plot(w/pi, ph); grid on;
xlabel('\omega/\pi'); ylabel('Unwrapped Phase (rad)');
title('Phase Response');

subplot(2,2,3);
zplane(z,p); grid on;
title('Pole-Zero Plot');

subplot(2,2,4);
stem(n, h, 'filled'); grid on;
xlabel('n'); ylabel('h[n]');
title('Impulse Response of the first 20 samples');

disp('First 20 samples of h[n]:');
disp(table(n(:), h(:), 'VariableNames', {'n','h'}));
```

4 Comparing Order of the Three Filter Design Methods

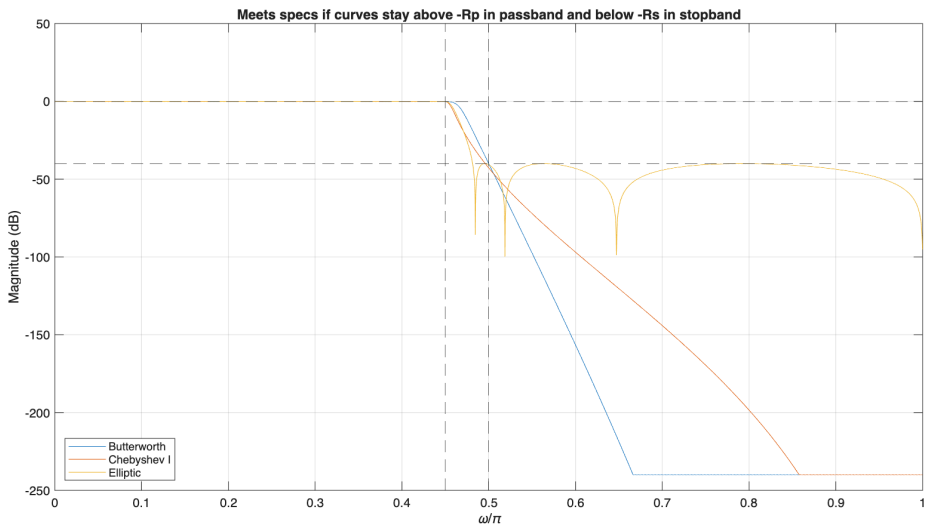


Figure 7: Plots generated

```
Butterworth: N = 42, Wn = 0.465168
Chebyshev I: N = 13, Wn = 0.450000
Elliptic   : N = 7, Wn = 0.450000

Lowest order meeting specs: Elliptic (N = 7)
```

Figure 8: First 20 values of the impulse response

Code & Explanation

Each function returns the required order N and the corresponding adjusted cut-off frequency Wn.

The code prints those orders, compares them, and selects the smallest one. For verification, it then designs each filter at its minimum order and plots their magnitude responses to visually confirm they satisfy the ripple and attenuation constraints.

Making sense of the results

- Butterworth: N = 42

- Chebyshev I: N = 13
- Elliptic: N = 7

This makes sense because the transition band is very narrow (from 0.45π to 0.5π), while the ripple requirements are strict (0.1 dB passband ripple and 40 dB stopband attenuation).

Butterworth filters have a maximally flat passband and no ripple anywhere, but that smoothness costs efficiency. They require very high order to achieve sharp transitions. Hence it needing an order 42.

Chebyshev Type I allows ripple in the passband, which gives it a sharper rolloff for the same order, so it only needs order 13.

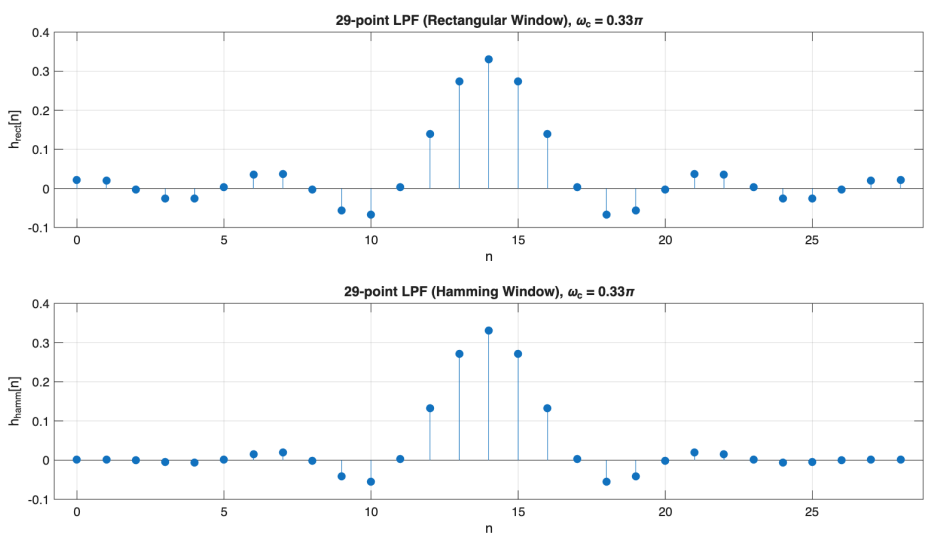
Elliptic filters allow ripple in both passband and stopband. Because they relax the smoothness requirement in both regions, they achieve the steepest transition band for a given order, which is why they meet the specs with only order 7.

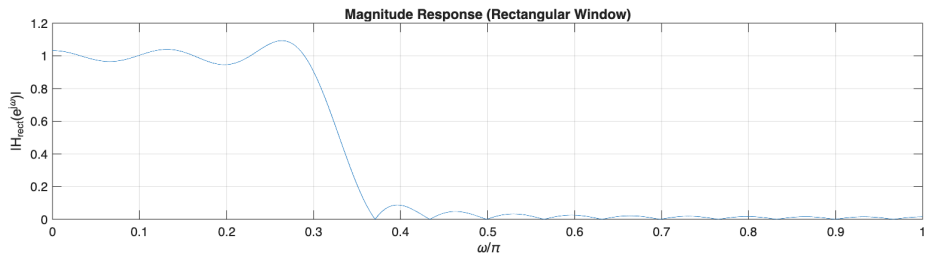
Hence it makes sense that Elliptic is the most efficient in terms of order, Butterworth is the most conservative/smooth, and Chebyshev sits in between.

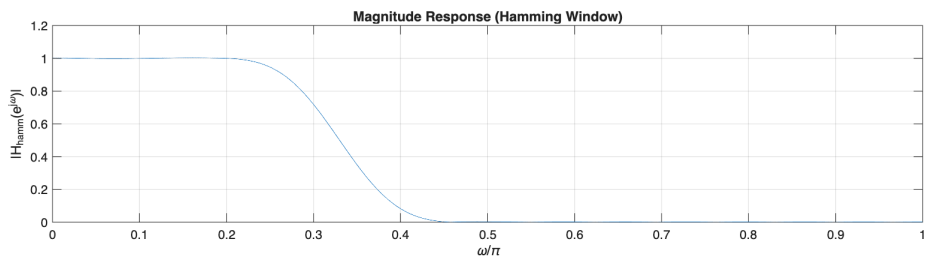
```
%% Question 4: Comparing the Order of the Three Filter Design Methods
% wp = 0.45pi, sb cutoff ws = 0.5pi, PB ripple = 0.1 dB, SB atten below 40
% dB. using buttord, cheb1ord, f200c ellipord commands.

clear; close all; clc;

wp = 0.45;
ws = 0.50;
Rp = 0.1;
Rs = 40;

%% Minimum order for each method
[Nb, Wnb] = buttord(wp, ws, Rp, Rs);      % Method 1: Butterworth
[Nc, Wnc] = cheb1ord(wp, ws, Rp, Rs);     % Method 2: Chebyshev Type I
[Ne, Wne] = ellipord(wp, ws, Rp, Rs);     % Method 3: Elliptic

fprintf('Butterworth: N = %d, Wn = %.6f\n', Nb, Wnb);
fprintf('Chebyshev I: N = %d, Wn = %.6f\n', Nc, Wnc);
fprintf('Elliptic   : N = %d, Wn = %.6f\n', Ne, Wne);

%% Finding the lowest-order method

Ns = [Nb Nc Ne];
names = {'Butterworth','Chebyshev I','Elliptic'};
[minN, idx] = min(Ns);
```

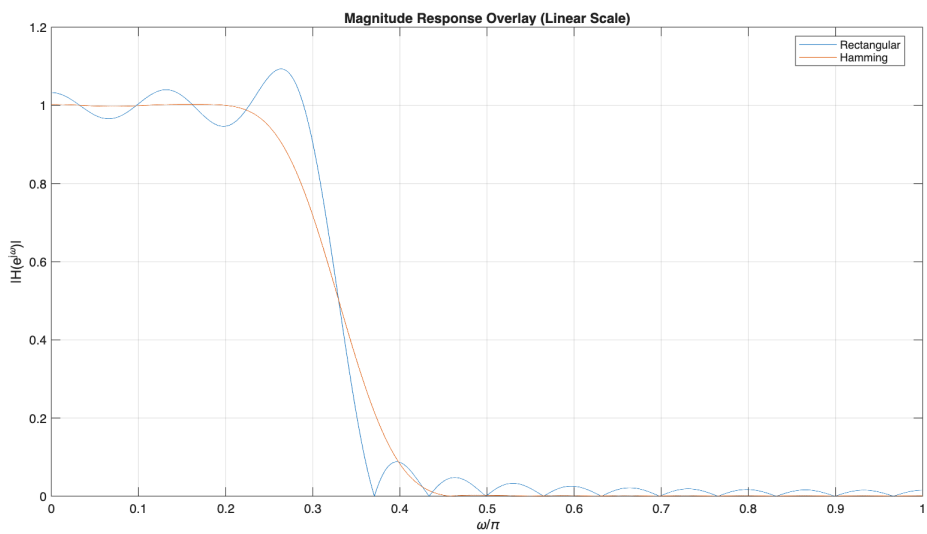
```
fprintf('\nLowest order meeting specs: %s (N = %d)\n', names{idx}, minN);

%% Method of verification: designing each filter at its minimum order
[bb,ab] = butter(Nb, Wnb, 'low');
[bc,ac] = cheby1(Nc, Rp, Wnc, 'low');
[be,ae] = ellip(Ne, Rp, Rs, Wne, 'low');

nfft = 4096;
[Hb,w] = freqz(bb,ab,nfft);
[Hc,~] = freqz(bc,ac,nfft);
[He,~] = freqz(be,ae,nfft);

figure('Name','Magnitude Responses','Color','w');
plot(w/pi, 20*log10(max(abs(Hb),1e-12))); hold on; grid on;
plot(w/pi, 20*log10(max(abs(Hc),1e-12)));
plot(w/pi, 20*log10(max(abs(He),1e-12)));
xline(wp,'--'); xline(ws,'--');
yline(-Rp,'--'); yline(-Rs,'--');
xlabel('\omega/\pi'); ylabel('Magnitude (dB)');
legend('Butterworth','Chebyshev I','Elliptic','Location','SouthWest');
title('Meets specs if curves stay above -Rp in passband and below -Rs in stopband');
```

5 Rectangular and Hamming Window Design









Rectangular Window (29-point FIR)

From the magnitude response plot:

$$\delta_{p1} = 1.09312$$

$$\delta_{p2} = 0.946991$$

$$\omega_p = 0.295776\,\pi$$

$$\delta_s = 0.0872631$$

$$\omega_s = 0.360718\,\pi$$

Hamming Window (29-point FIR)

From the magnitude response plot:

$$\delta_{p1} = 1.00143$$

$$\delta_{p2} = 0.997536$$

$$\omega_p = 0.20813\,\pi$$

$$\delta_s = 0.00281059$$

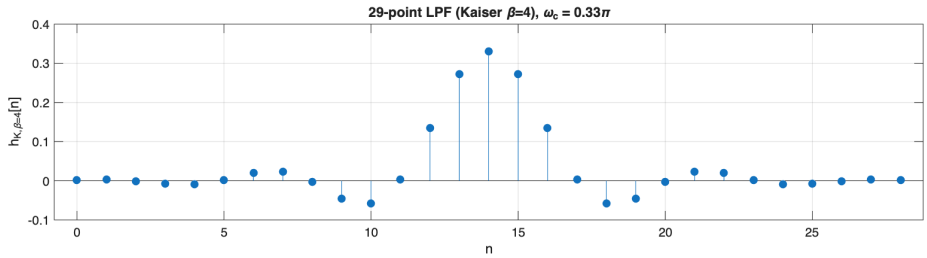
$$\omega_s = 0.449829\,\pi$$

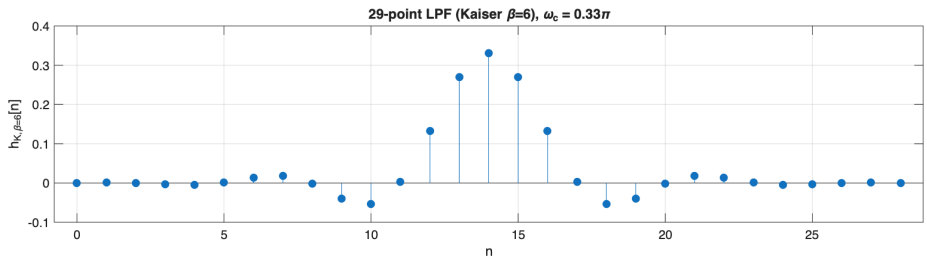
Code

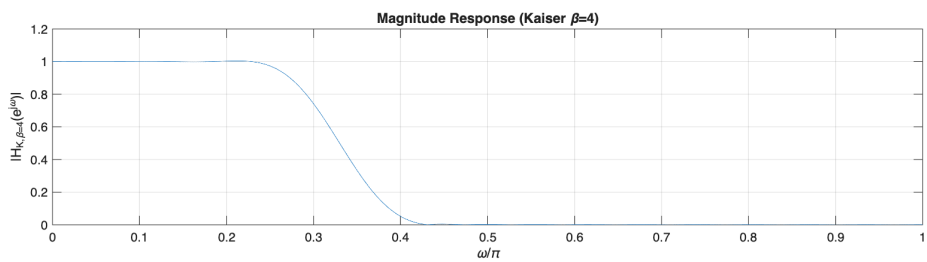
```
%% Question 5: Rectangular & Hamming
clear; close all; clc;

%% Specs
N  = 29;
M  = (N-1)/2;
wc = 0.33*pi;

%% Ideal LPF impulse response (shifted sinc)
n  = 0:N-1;
hd = sin(wc*(n - M)) ./ (pi*(n - M));
hd(n==M) = wc/pi;

%% Windows
w_rect = ones(N,1);
w_hamm = hamming(N);

%% FIR coefficients
h_rect = hd(:).*w_rect;
h_hamm = hd(:).*w_hamm;
```

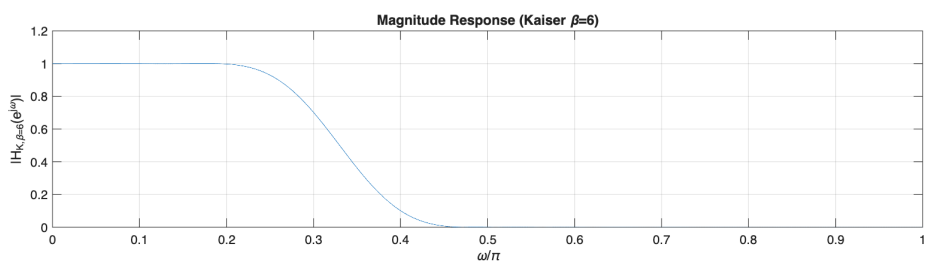
```
%% Frequency response
Nfft = 8192;
[Hrect,w] = freqz(h_rect,1,Nfft);
[Hhamm,~] = freqz(h_hamm,1,Nfft);

%% Plots for the impulse response
figure('Name','Impulse Responses','Color','w');
subplot(2,1,1);
stem(n, h_rect, 'filled'); grid on;
xlabel('n'); ylabel('h_{rect}[n]');
title(sprintf('29-point LPF (Rectangular Window), \\omega_c = %.2f\\pi', wc/pi));

subplot(2,1,2);
stem(n, h_hamm, 'filled'); grid on;
xlabel('n'); ylabel('h_{hamm}[n]');
title(sprintf('29-point LPF (Hamming Window), \\omega_c = %.2f\\pi', wc/pi));

%% Plots for the Magnitude reponses
figure('Name','Magnitude Responses','Color','w');
subplot(2,1,1);
plot(w/pi, abs(Hrect)); grid on; xlim([0 1]);
xlabel('\omega/\pi'); ylabel('|H_{rect}(e^{j\omega})|');
title('Magnitude Response (Rectangular Window)');

subplot(2,1,2);
plot(w/pi, abs(Hhamm)); grid on; xlim([0 1]);
xlabel('\omega/\pi'); ylabel('|H_{hamm}(e^{j\omega})|');
title('Magnitude Response (Hamming Window)');

%% For visual understanding: overlay
figure('Name','Magnitude Response Overlay','Color','w');
plot(w/pi, abs(Hrect)); hold on;
plot(w/pi, abs(Hhamm));
grid on; xlim([0 1]);
xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
legend('Rectangular','Hamming','Location','best');
title('Magnitude Response Overlay (Linear Scale)');
```

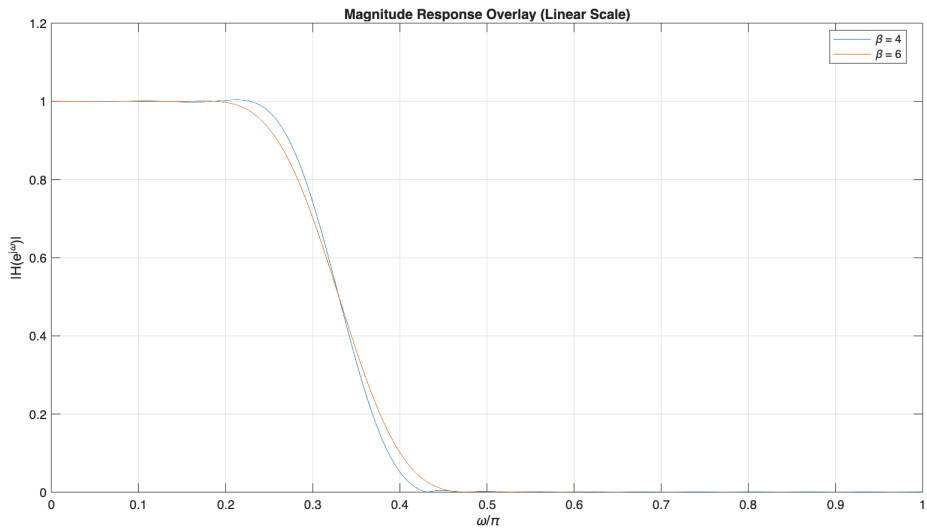
6 Kaiser Window Design











Kaiser Window ($\beta = 4$)

From the magnitude response plot:

$$\delta_{p1} = 1.00396$$

$$\delta_{p2} = 0.99843$$

$$\omega_p = 0.231079\,\pi$$

$$\delta_s = 0.00406489$$

$$\omega_s = 0.425415\,\pi$$

Kaiser Window ($\beta = 6$)

From the magnitude response plot:

$$\delta_{p1} = 1.00000$$

$$\delta_{p2} = 0.999654$$

$$\omega_p = 0.189575\,\pi$$

$$\delta_s = 0.000811513$$

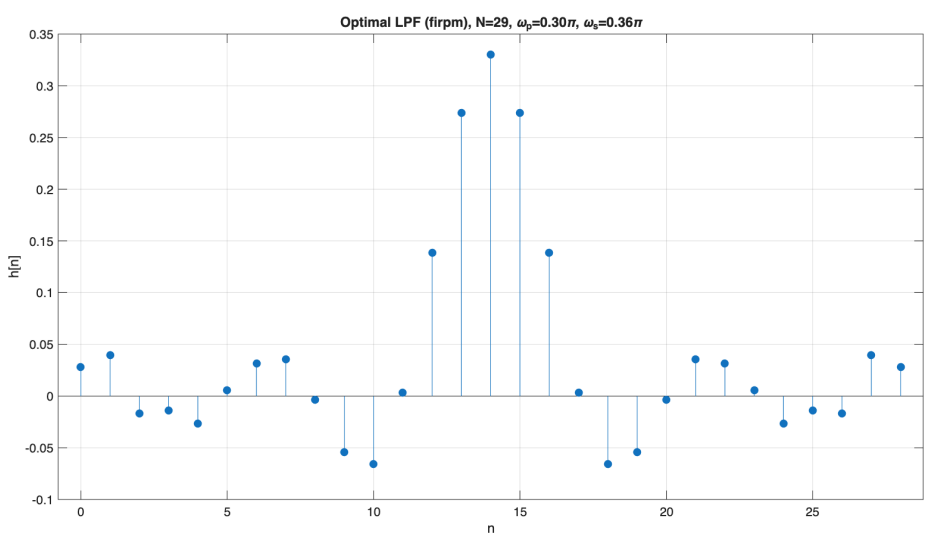
$$\omega_s = 0.466675\,\pi$$

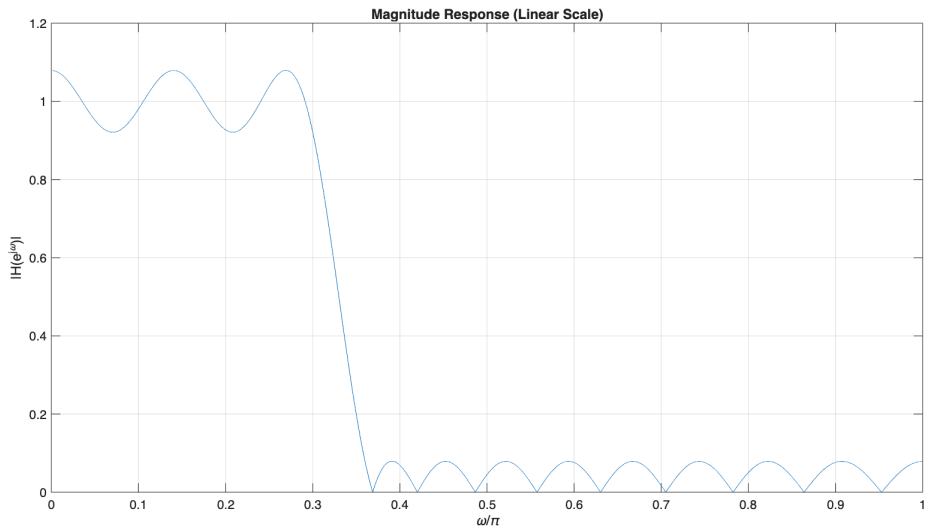
Code

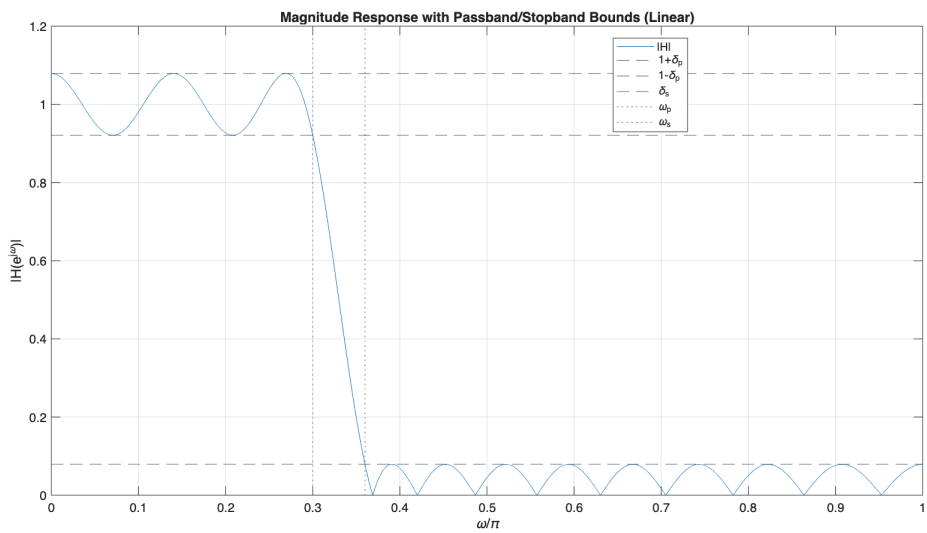
```
%% Question 6 Kaiser Window FIR Lowpass (beta = 4 and 6)
clear; close all; clc;

%% Specs
N  = 29;
M  = (N-1)/2;
wc = 0.33*pi;

%% Ideal LPF impulse response (shifted sinc)
n  = 0:N-1;
hd = sin(wc*(n - M)) ./ (pi*(n - M));
hd(n==M) = wc/pi;

%% Kaiser windows
beta1 = 4;
beta2 = 6;
w_k4  = kaiser(N, beta1);
w_k6  = kaiser(N, beta2);

%% FIR coefficients
h_k4 = hd(:).*w_k4;
h_k6 = hd(:).*w_k6;

%% Frequency response
Nfft = 8192;
[Hk4, w] = freqz(h_k4, 1, Nfft);
[Hk6, ~] = freqz(h_k6, 1, Nfft);

%% Impulse responses plots
figure('Name','Impulse Responses (Kaiser)','Color','w');
subplot(2,1,1);
stem(n, h_k4, 'filled'); grid on;
xlabel('n'); ylabel('h_{K,\beta=4}[n]');
title(sprintf('29-point LPF (Kaiser \\beta=%g), \\omega_c = %.2f\\pi', beta1, wc/pi));

subplot(2,1,2);
stem(n, h_k6, 'filled'); grid on;
xlabel('n'); ylabel('h_{K,\beta=6}[n]');
```

```
title(sprintf('29-point LPF (Kaiser \\beta=%g), \\omega_c = %.2f\\pi', beta2, wc/pi));

%% Magnitude Response plots
figure('Name','Magnitude Responses (Kaiser)','Color','w');
subplot(2,1,1);
plot(w/pi, abs(Hk4)); grid on; xlim([0 1]);
xlabel('\omega/\pi'); ylabel('|H_{K,\beta=4}(e^{j\omega})|');
title('Magnitude Response (Kaiser \beta=4)');

subplot(2,1,2);
plot(w/pi, abs(Hk6)); grid on; xlim([0 1]);
xlabel('\omega/\pi'); ylabel('|H_{K,\beta=6}(e^{j\omega})|');
title('Magnitude Response (Kaiser \beta=6)');

figure('Name','Magnitude Response Overlay (Kaiser)','Color','w');
plot(w/pi, abs(Hk4)); hold on;
plot(w/pi, abs(Hk6));
grid on; xlim([0 1]);
xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
legend('\beta = 4','\beta = 6','Location','best');
title('Magnitude Response Overlay (Linear Scale)');
```

7 Optimal Low Pass Filter




Magnitude Response (Linear Scale)
|H(e^{jω})|
ω/π
0
0.2
0.4
0.6
0.8
1
1.2
0.1
0.3
0.5
0.7
0.9


Magnitude Response with Passband/Stopband Bounds (Linear)
|H|
1+δp
1-δp
δs
ωp
ωs
|H(e^{jω})|
ω/π
0
0.2
0.4
0.6
0.8
1
1.2
0.1
0.3
0.5
0.7
0.9

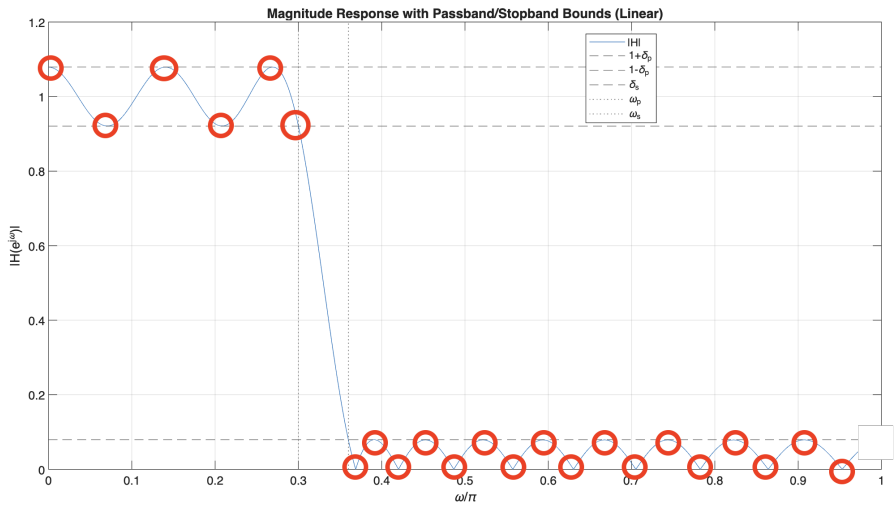


Figure 9: Alternation points circled

Ripple Values

$$\delta_{p1} = 1.0793324$$

$$\delta_{p2} = 0.92107247$$

I calculated the passband ripple as:

$$\delta_p = \max_{\omega \leq \omega_p} \Big| \, |H(e^{j\omega})| - 1 \, \Big| = 0.079332377.$$

and the stopband ripple as:

$$\delta_s = \max_{\omega \geq \omega_s} |H(e^{j\omega})| = 0.079404062.$$

Thus, in the passband:

$$1 - \delta_p \leq |H(e^{j\omega})| \leq 1 + \delta_p,$$

with

$$1 - \delta_p = 0.920667623, \qquad 1 + \delta_p = 1.079332377.$$

In the stopband:

$$|H(e^{j\omega})| \leq \delta_s = 0.079404062.$$

Alternation Points

By visually circling the extrema on the magnitude response, I observed a total of

23 alternation points.

Out of these,

5 occurred in the passband

and

18 occurred in the stopband.

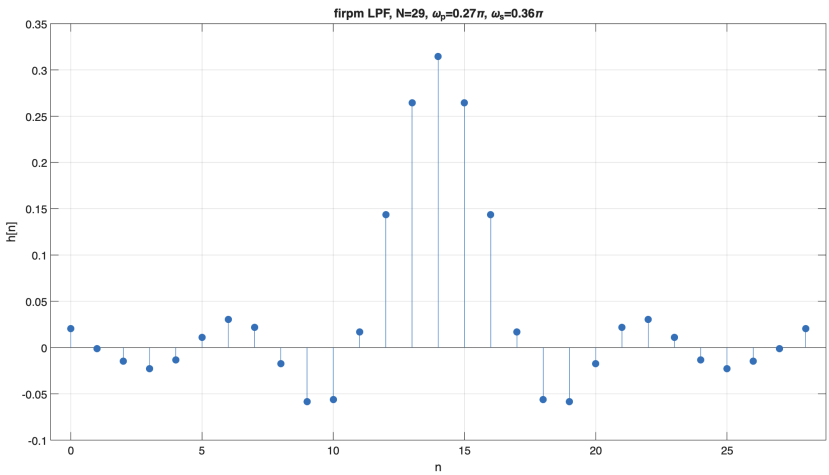
Hence I found that the error alternated between positive and negative extrema of nearly equal magnitude in both bands, which confirmed that the filter exhibited equiripple behavior consistent with Chebyshev optimality.

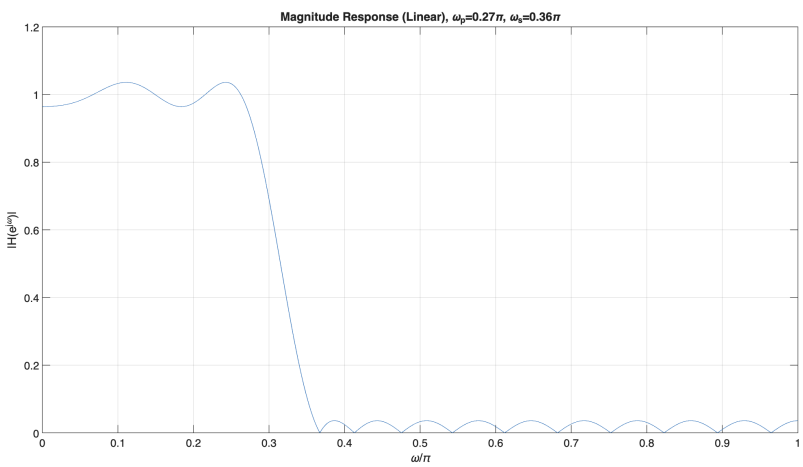
Code

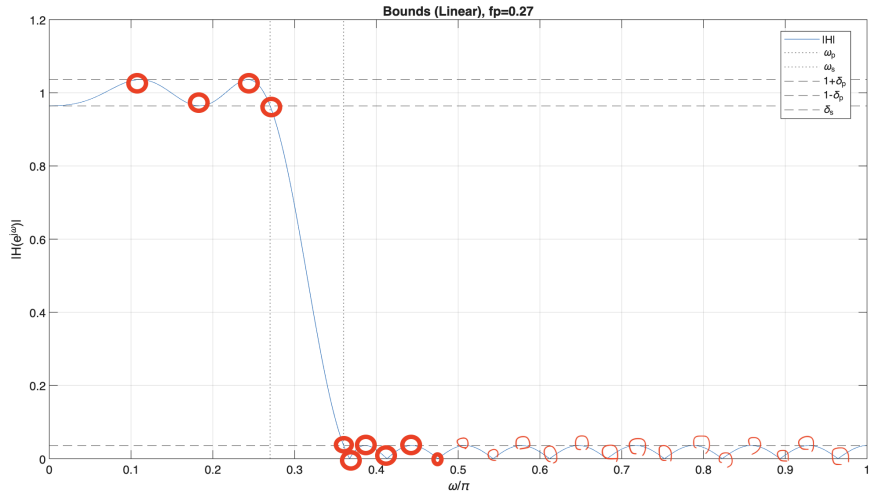
```
%% Question 7: Optimal Lowpass Filter
clear; close all; clc;

%% Specs
N  = 29;              % length
ord = N-1;            % firpm order
wp = 0.30;            % passband edge
ws = 0.36;            % stopband edge

%% Design (equal weighting in passband and stopband)
h = firpm(ord, [0 wp ws 1], [1 1 0 0]);   % freq grid normalized to [0,1] == [0,pi]

%% Frequency response
Nfft = 8192;
[H,w] = freqz(h, 1, Nfft);
mag = abs(H);
wn  = w/pi;

%% Plots for impulse response
n = 0:ord;
figure('Name','Impulse Response (firpm)','Color','w');
stem(n, h, 'filled'); grid on;
xlabel('n'); ylabel('h[n]');
title(sprintf('Optimal LPF (firpm), N=%d,
\\omega_p=%.2f\\pi, \\omega_s=%.2f\\pi', N, wp, ws));

%% Plots for magnitude response
figure('Name','Magnitude Response (firpm)','Color','w');
plot(wn, mag); grid on; xlim([0 1]);
xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
```

```
title('Magnitude Response (Linear Scale)');

%% Ripple limits
pb = (wn <= wp);          % passband region
sb = (wn >= ws);          % stopband region

delta_p1 = max(mag(pb));  % max magnitude in pb
delta_p2 = min(mag(pb));  % min magnitude in pb

%% Requested deltas:
% 1 - δp <= |H| <= 1 + δp in passband  => δp = max(|H-1|) over passband
delta_p = max(abs(mag(pb) - 1));

%% δs >= |H| in stopband => δs = max(|H|) over stopband
delta_s = max(mag(sb));

fprintf('\nExercise 7 (firpm) Ripple Results\n');
fprintf('Passband max |H| (delta_p1) = %.8g\n', delta_p1);
fprintf('Passband min |H| (delta_p2) = %.8g\n', delta_p2);
fprintf('delta_p (max |H-1| in passband) = %.8g\n', delta_p);
fprintf('delta_s (max |H| in stopband)   = %.8g\n', delta_s);

figure('Name','Magnitude Response w/ Bounds','Color','w');
plot(wn, mag); grid on; xlim([0 1]);
xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
title('Magnitude Response with Passband/Stopband Bounds (Linear)');
hold on;
yline(1 + delta_p, '--');
yline(1 - delta_p, '--');
yline(delta_s,     '--');
xline(wp, ':'); xline(ws, ':');
legend('|H|','1+\delta_p','1-\delta_p','\delta_s','\omega_p',
'\omega_s','Location','best');
```

8 Varying the Passband Edge







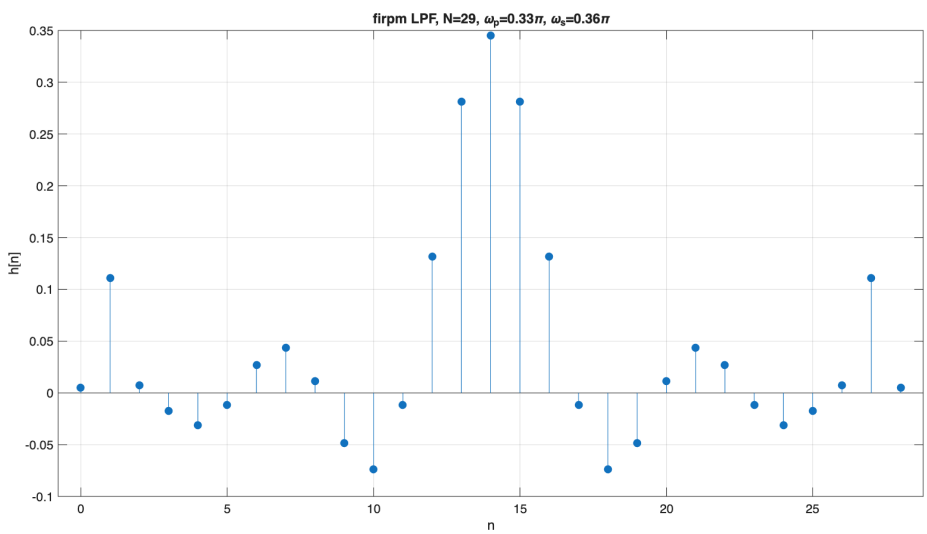
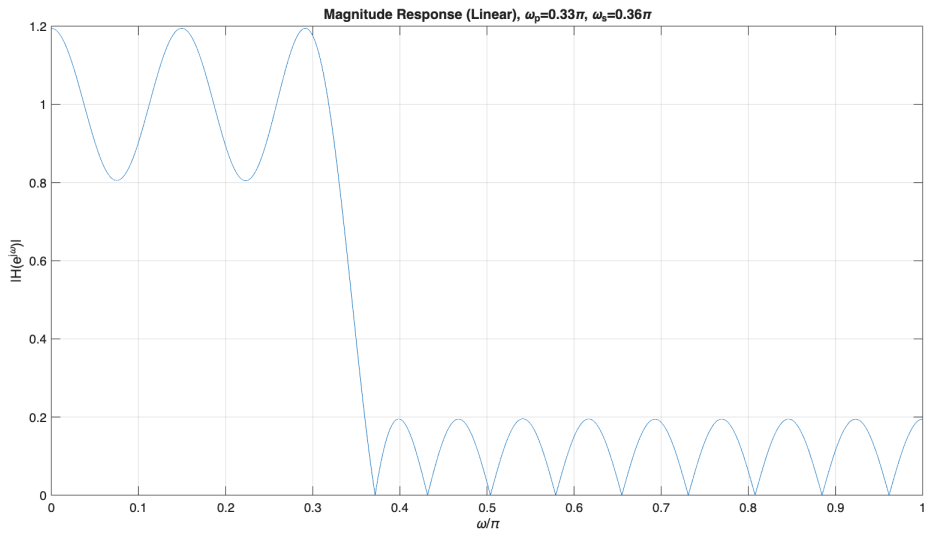
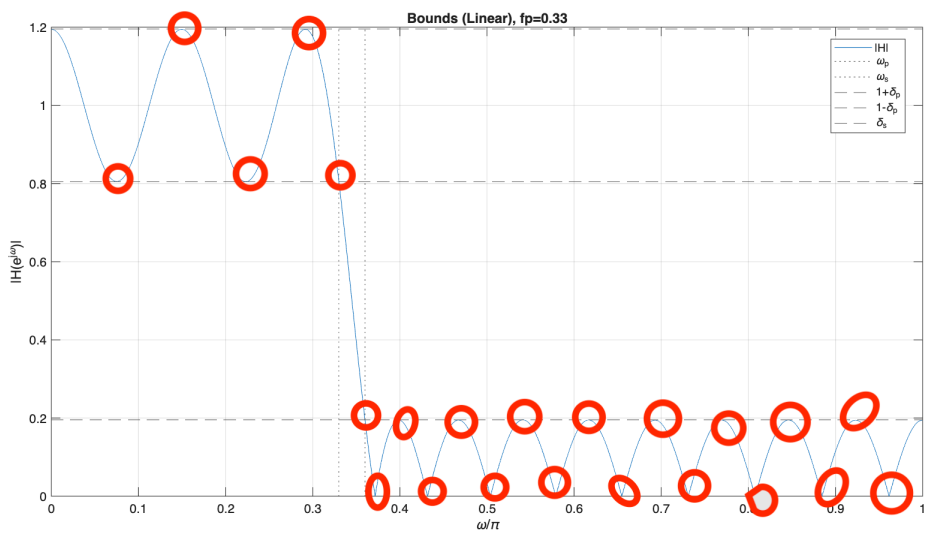


Figure 10: Alternation points circled





Case 1: $f_p = 0.27$

From the magnitude response, I obtained

$$\delta_{p1} = 1.0359509$$

$$\delta_{p2} = 0.96414324$$

Passband ripple was:

$$\delta_p = 0.035950911.$$

Stopband ripple was:

$$\delta_s = 0.036053164.$$

$$1 - \delta_p = 0.964049089, \qquad 1 + \delta_p = 1.035950911,$$

and in the stopband,

$$|H(e^{j\omega})| \leq 0.036053164.$$

By visually circling the extrema, I counted

$$\text{24 alternation points.}$$

Out of these,

$$\text{4 occurred in the passband}$$

and

$$\text{20 occurred in the stopband.}$$

Case 2: $f_p = 0.33$

From the magnitude response, I obtained

$$\delta_{p1} = 1.1946703$$
$$\delta_{p2} = 0.80462368$$

I calculated the passband ripple as

$$\delta_p = 0.19537632.$$

and the stopband ripple as

$$\delta_s = 0.19538283.$$

$$1 - \delta_p = 0.80462368, \qquad 1 + \delta_p = 1.19537632,$$

and in the stopband,

$$|H(e^{j\omega})| \leq 0.19538283.$$

I counted

$$\text{23 alternation points.}$$

Out of these,

$$\text{5 occurred in the passband}$$

and

$$\text{18 occurred in the stopband.}$$

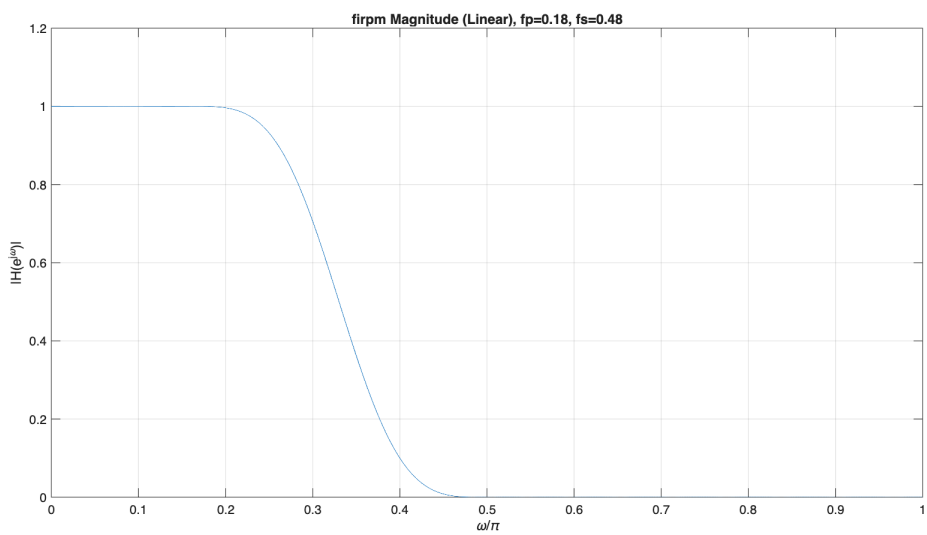
It seems that increasing the passband edge significantly increased both δ_p and δ_s, and the distribution of alternation points shifted toward a more balanced allocation between passband and stopband.

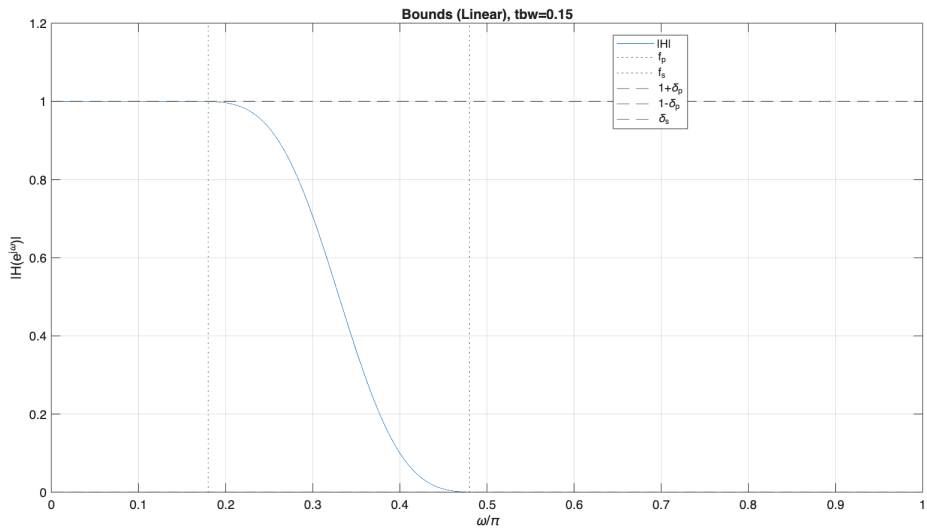
Code

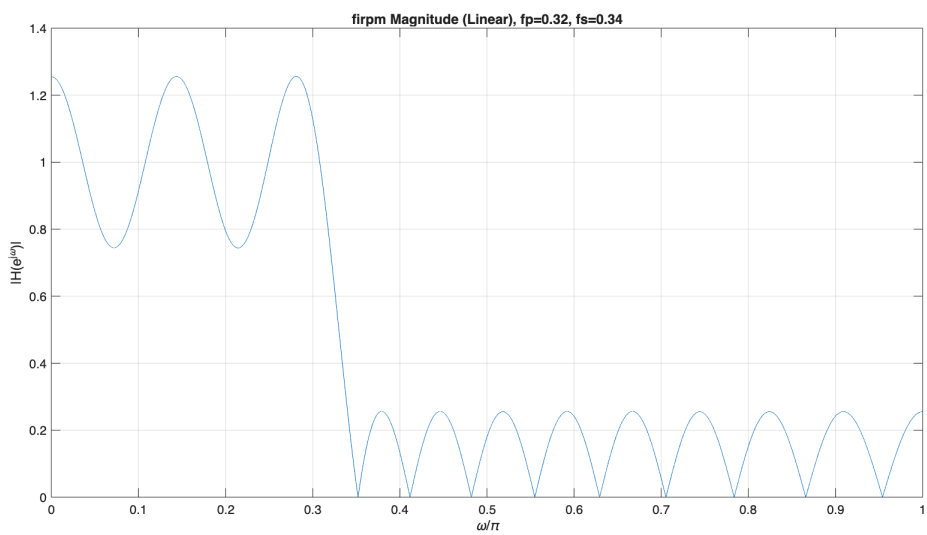
```
%% Question 8: Varying the Passband Edge (firpm), fp = 0.27 and 0.33
clear; close all; clc;

%% Specs
N   = 29;
ord = N-1;
ws  = 0.36;
fps = [0.27 0.33];
Nfft = 8192;

for k = 1:numel(fps)
    fp = fps(k);

    %% Design (equal weighting in passband and stopband)
    h = firpm(ord, [0 fp ws 1], [1 1 0 0]);

    %% Frequency response
    [H,w] = freqz(h, 1, Nfft);
    mag = abs(H);
    wn  = w/pi;

    %% Ripple computations (NOT dB)
    pb = (wn <= fp);
    sb = (wn >= ws);

    delta_p1 = max(mag(pb));
    delta_p2 = min(mag(pb));
    delta_p  = max(abs(mag(pb) - 1));
    delta_s  = max(mag(sb));

    fprintf('\nfp = %.2f\n', fp);
    fprintf('delta_p1 = %.8g\n', delta_p1);
    fprintf('delta_p2 = %.8g\n', delta_p2);
    fprintf('delta_p  = %.8g\n', delta_p);
    fprintf('delta_s  = %.8g\n', delta_s);

    %% Impulse response plot
    n = 0:ord;
    figure('Color','w','Name',sprintf('Impulse Response (fp=%.2f)',fp));
    stem(n, h, 'filled'); grid on;
    xlabel('n'); ylabel('h[n]');
    title(sprintf('firpm LPF, N=%d, \\omega_p=%.2f\\pi, \\omega_s=%.2f\\pi', N, fp, ws));

    %% Magnitude response plot
```

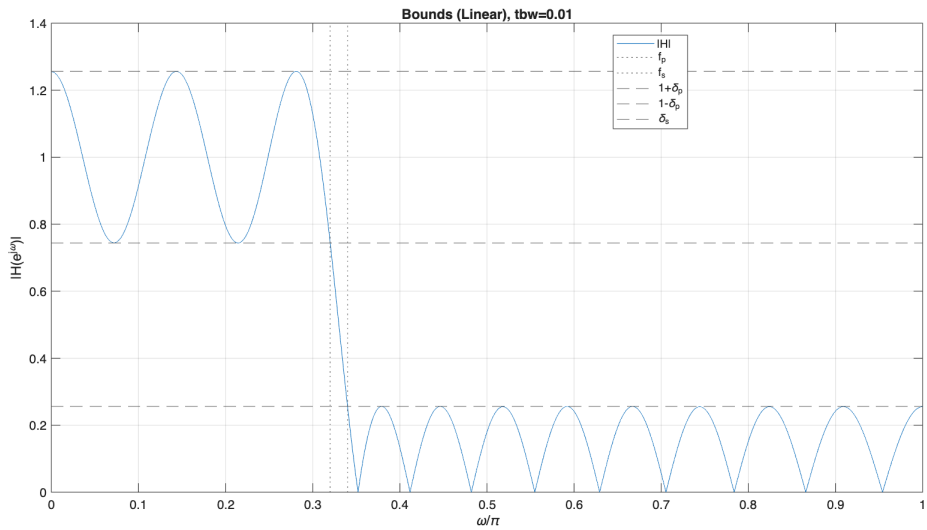
```
    figure('Color','w','Name',sprintf('Magnitude Response (fp=%.2f)',fp));
    plot(wn, mag); grid on; xlim([0 1]);
    xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
    title(sprintf('Magnitude Response (Linear),
    \\omega_p=%.2f\\pi, \\omega_s=%.2f\\pi', fp, ws));

    %% Bounds overlay (for when I circle the alternation points)
    figure('Color','w','Name',sprintf('Magnitude w/ Bounds (fp=%.2f)',fp));
    plot(wn, mag); grid on; xlim([0 1]); hold on;
    xline(fp, ':'); xline(ws, ':');
    yline(1 + delta_p, '--');
    yline(1 - delta_p, '--');
    yline(delta_s, '--');
    xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
    title(sprintf('Bounds (Linear), fp=%.2f', fp));
    legend('|H|','\omega_p','\omega_s','1+\delta_p',
    '1-\delta_p','\delta_s','Location','best');
end
```

9 Varying the Transition Bandwidth




Bounds (Linear), tbw=0.15
|H|
f_p
f_s
$1+\delta_p$
$1-\delta_p$
δ_s
$|H(e^{j\omega})|$
ω/π


firpm Magnitude (Linear), fp=0.32, fs=0.34
$|H(e^{j\omega})|$
ω/π



Case 1: tbw $= 0.15$

In this case,

$$f_p = 0.18, \qquad f_s = 0.48.$$

From the magnitude response, I obtained

$$\delta_p = 0.00014957642,$$

$$\delta_s = 0.00014893384.$$

Thus,

$$1 - \delta_p \leq |H(e^{j\omega})| \leq 1 + \delta_p \quad \text{for } \omega \leq 0.18\pi,$$

and

$$|H(e^{j\omega})| \leq 0.00014893384 \quad \text{for } \omega \geq 0.48\pi.$$

I observed that with a wide transition band, both δ_p and δ_s became extremely small, indicating very low ripple.

Case 2: tbw $= 0.01$

In this case,

$$f_p = 0.32, \qquad f_s = 0.34.$$

From the magnitude response, I obtained

$$\delta_p = 0.25620166,$$

$$\delta_s = 0.25606341.$$

Thus,

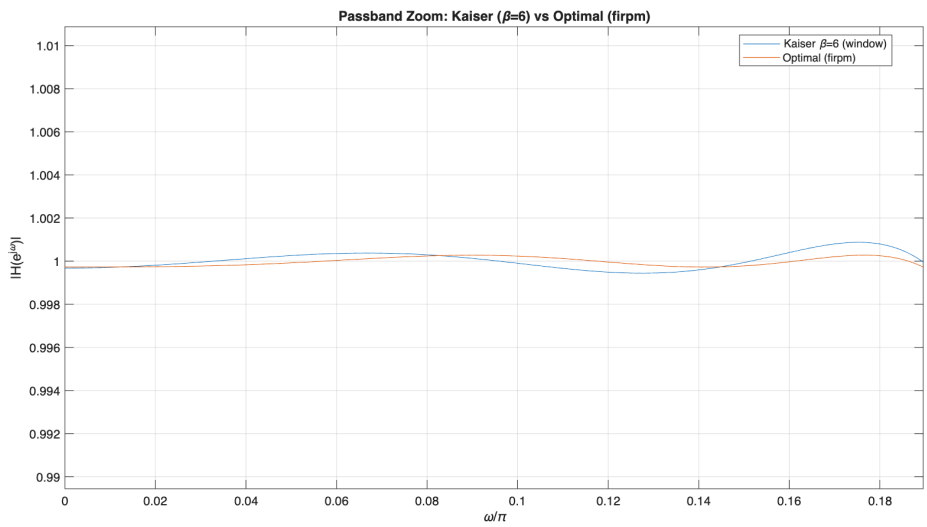
$$1 - \delta_p \leq |H(e^{j\omega})| \leq 1 + \delta_p \quad \text{for } \omega \leq 0.32\pi,$$

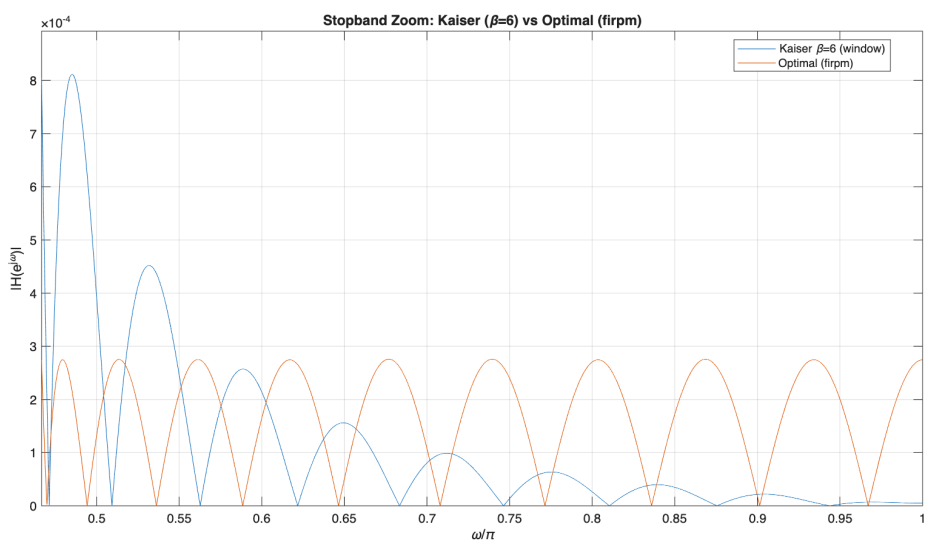
and

$$|H(e^{j\omega})| \leq 0.25606341 \quad \text{for } \omega \geq 0.34\pi.$$

I observed that when the transition bandwidth became very narrow, both δ_p and δ_s increased dramatically.

10 Optimal vs. Window FIR Design





Using `firpm` with the same filter length ($N = 29$), the same passband edge ω_p, and the same stopband edge ω_s as the Kaiser $\beta = 6$ design, I got the following ripple values for the optimal equiripple filter:

$$\delta_p = 0.00027951396$$

$$\delta_s = 0.00027580727$$

For comparison, the Kaiser window ($\beta = 6$) design produced:

$$\delta_p = 0.00087531788, \qquad \delta_s = 0.00081151727.$$

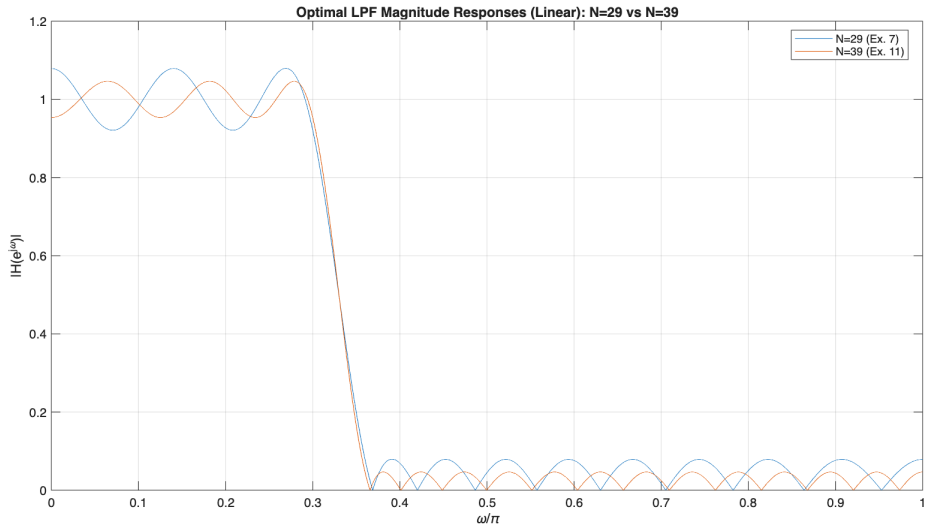
Bboth δ_p and δ_s for the optimal filter were significantly smaller than those of the window-method filter. Therefore, it seems like the the optimal (equiripple) design performed better than the Kaiser window design for the same filter length and band edges, since it achieved smaller passband and stopband ripples.

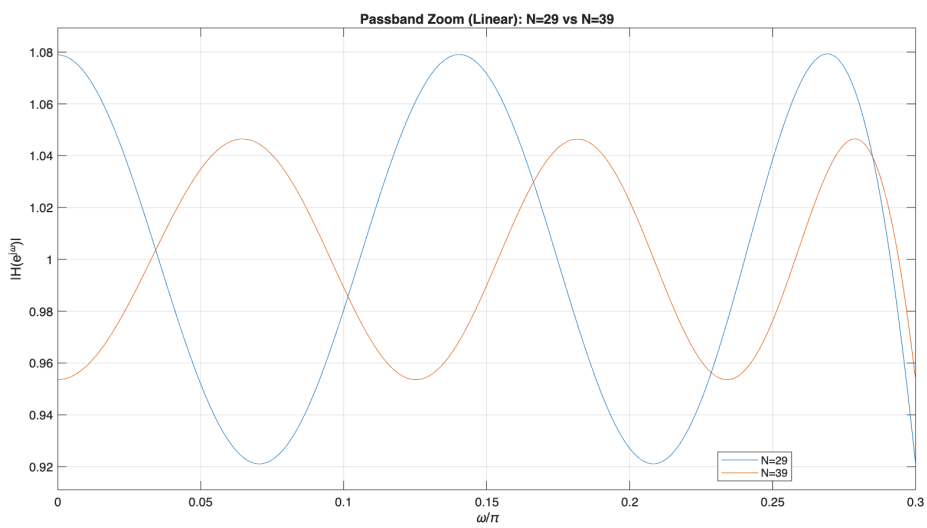
Code

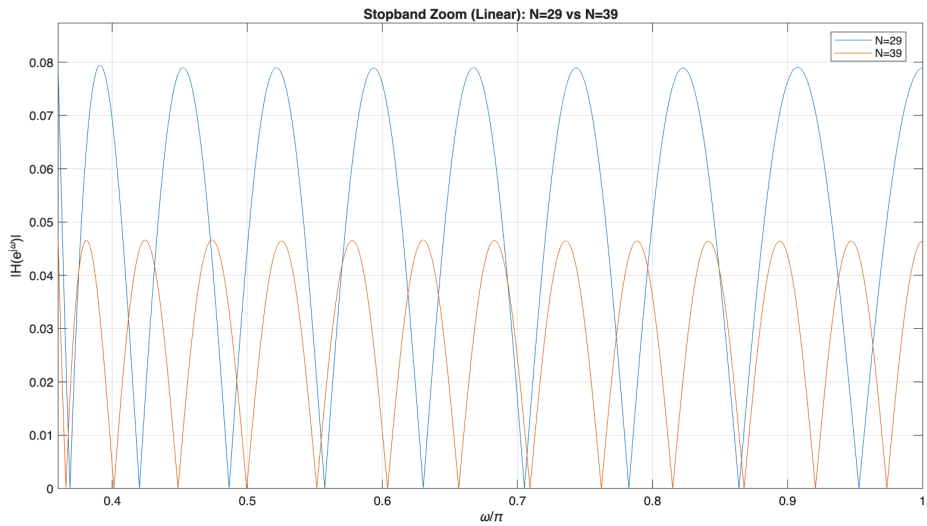
```
%% Exercise 10: Optimal vs. Window FIR
clear; close all; clc;

% Given from Exercise 6 (Kaiser beta=6)
wp = 0.189575;
ws = 0.466675;

% Specs
N    = 29;
ord  = N-1;
wc   = 0.33*pi;        % same ideal cutoff used in ex. 6
beta = 6;
Nfft = 8192;

%% Window-method filter: Kaiser beta=6
n  = 0:ord;
M  = ord/2;
hd = sin(wc*(n - M)) ./ (pi*(n - M));
hd(n==M) = wc/pi;

wK = kaiser(N, beta).';
h_win = (hd .* wK).';          % column

%% Optimal equiripple filter with same wp, ws
h_opt = firpm(ord, [0 wp ws 1], [1 1 0 0]);

%% Frequency responses
```

```
[Hwin, w] = freqz(h_win, 1, Nfft);
[Hopt, ~] = freqz(h_opt, 1, Nfft);
magW = abs(Hwin);
magO = abs(Hopt);
wn   = w/pi;

%% Compute deltas
pb = (wn <= wp);
sb = (wn >= ws);

delta_p_opt = max(abs(magO(pb) - 1));
delta_s_opt = max(magO(sb));

delta_p_win = max(abs(magW(pb) - 1));
delta_s_win = max(magW(sb));

fprintf('\nOptimal (firpm): delta_p = %.8g, delta_s = %.8g\n', delta_p_opt, delta_s_opt);
fprintf('Window (Kaiser beta=6): delta_p = %.8g,
delta_s = %.8g\n', delta_p_win, delta_s_win);

%% Plot 1: zoom in on passband
figure('Color','w','Name','Passband Zoom');
plot(wn, magW, wn, magO); grid on;
xlim([0 wp]);
ylim([min([magW(pb); magO(pb)])-0.01, max([magW(pb); magO(pb)])+0.01]);
xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
title('Passband Zoom: Kaiser (\beta=6) vs Optimal (firpm)');
legend('Kaiser \beta=6 (window)','Optimal (firpm)','Location','best');

%% Plot 2: zoom in on stopband
figure('Color','w','Name','Stopband Zoom');
plot(wn, magW, wn, magO); grid on;
xlim([ws 1]);
ylim([0, max([magW(sb); magO(sb)])*1.1]);
xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
title('Stopband Zoom: Kaiser (\beta=6) vs Optimal (firpm)');
legend('Kaiser \beta=6 (window)','Optimal (firpm)','Location','best');
```

11 Changing Filter Length







$$\delta_p = 0.046497471,$$
$$\delta_s = 0.046612781.$$

Thus, in the passband,

$$1 - \delta_p \leq |H(e^{j\omega})| \leq 1 + \delta_p \quad \text{for } \omega \leq 0.30\pi,$$

and in the stopband,

$$|H(e^{j\omega})| \leq 0.046612781 \quad \text{for } \omega \geq 0.36\pi.$$

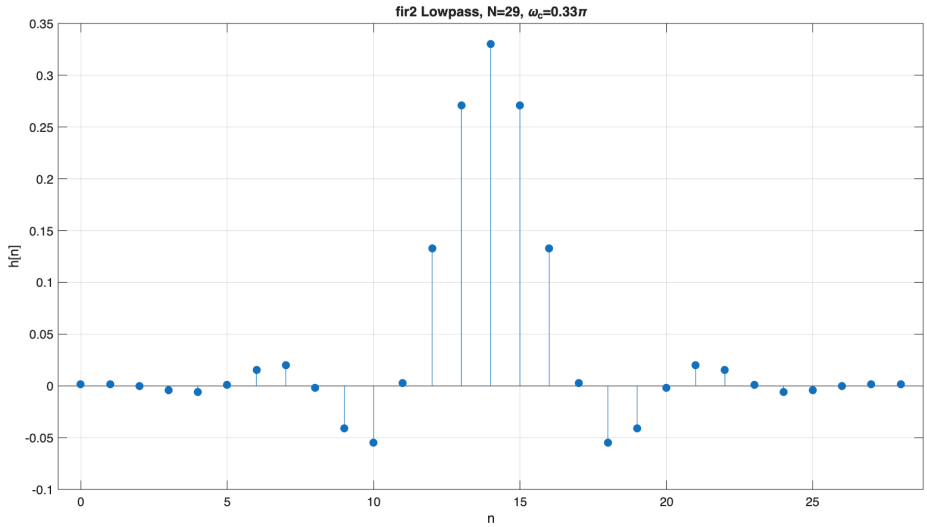
I concluded that increasing the filter length from 29 to 39 reduced both δ_p and δ_s, indicating improved ripple performance for the longer optimal filter.

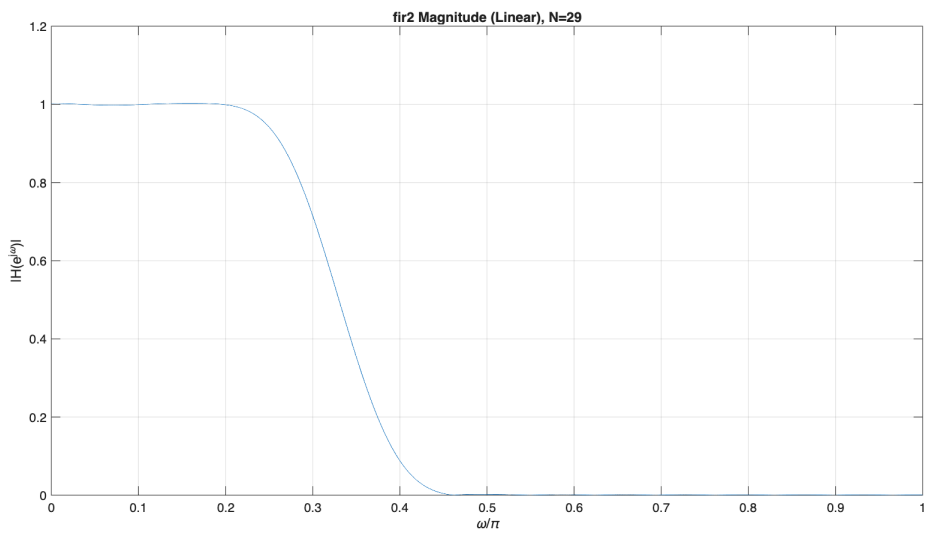
Code

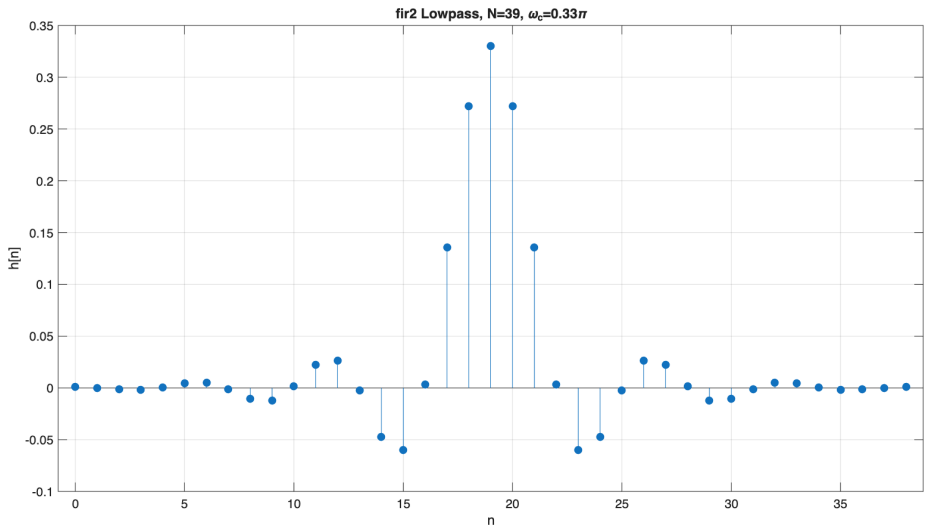
```
%% Exercise 11: Changing Filter Length
% (firpm) | N=39 vs N=29 (from ex. 7)
clear; close all; clc;

%% Specs (shoudl be same band edges as exercise 7)
wp = 0.30;
ws = 0.36;
Nfft = 8192;

%% Optimal 29-point filter
N29   = 29;
ord29 = N29 - 1;
h29   = firpm(ord29, [0 wp ws 1], [1 1 0 0]);
```

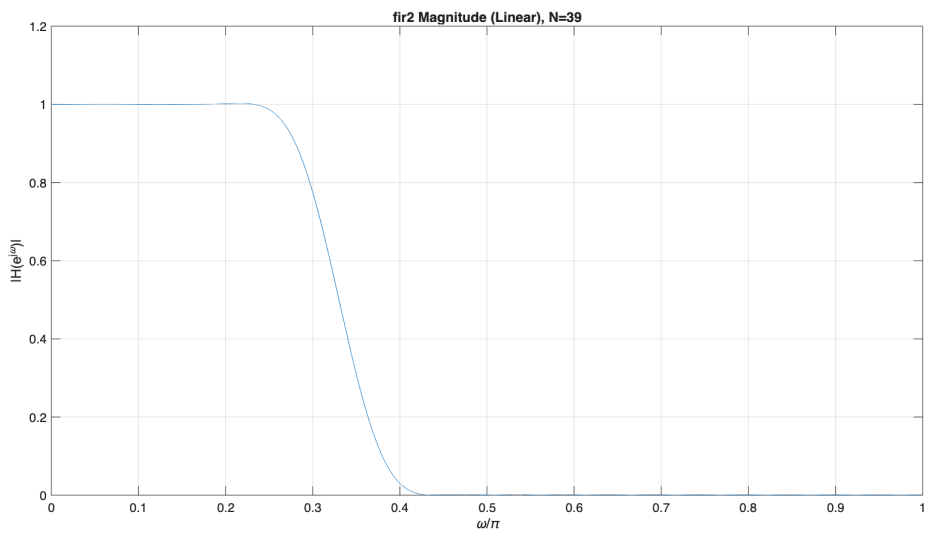
```
[H29,w] = freqz(h29, 1, Nfft);
mag29   = abs(H29);
wn      = w/pi;

%% Optimal 39-point filter
N39   = 39;
ord39 = N39 - 1;
h39   = firpm(ord39, [0 wp ws 1], [1 1 0 0]);

[H39,~] = freqz(h39, 1, Nfft);
mag39   = abs(H39);

%% Ripples for the 39-point filter
pb = (wn <= wp);
sb = (wn >= ws);

delta_p_39 = max(abs(mag39(pb) - 1));
delta_s_39 = max(mag39(sb));

fprintf('\n39-point optimal (firpm):\n');
fprintf('delta_p = %.8g\n', delta_p_39);
fprintf('delta_s = %.8g\n', delta_s_39);

%% Both mag responses plot
figure('Color','w','Name','Magnitude Response: N=29 vs N=39');
plot(wn, mag29, wn, mag39); grid on; xlim([0 1]);
xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
title('Optimal LPF Magnitude Responses (Linear): N=29 vs N=39');
legend('N=29 (Ex. 7)','N=39 (Ex. 11)','Location','best');

%% Zoomed plots (so i can visually compare ripple reduction)
figure('Color','w','Name','Passband Zoom: N=29 vs N=39');
plot(wn, mag29, wn, mag39); grid on;
xlim([0 wp]);
ylim([min([mag29(pb); mag39(pb)])-0.01, max([mag29(pb); mag39(pb)])+0.01]);
xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
title('Passband Zoom (Linear): N=29 vs N=39');
legend('N=29','N=39','Location','best');

figure('Color','w','Name','Stopband Zoom: N=29 vs N=39');
plot(wn, mag29, wn, mag39); grid on;
xlim([ws 1]);
ylim([0, max([mag29(sb); mag39(sb)])*1.1]);
xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
title('Stopband Zoom (Linear): N=29 vs N=39');
legend('N=29','N=39','Location','best');
```

12 Comparison of filters









Case 1: $N = 29$

$$\delta_{p1} = 1.00278, \qquad \delta_{p2} = 0.997819,$$

$$\omega_p = 0.203979\,\pi,$$

$$\delta_s = 0.00271975, \qquad \omega_s = 0.453735\,\pi.$$

Thus, in the passband,

$$1 - \delta_p \leq |H(e^{j\omega})| \leq 1 + \delta_p,$$

where $\delta_p \approx 0.00278$, and in the stopband,

$$|H(e^{j\omega})| \leq 0.00271975.$$

Case 2: $N = 39$

$$\delta_{p1} = 1.00196, \qquad \delta_{p2} = 0.999603,$$

$$\omega_p = 0.234131\,\pi,$$

$$\delta_s = 0.00144913, \qquad \omega_s = 0.426514\,\pi.$$

Thus, in the passband,

$$\delta_p \approx 0.00196,$$

and in the stopband,

$$|H(e^{j\omega})| \leq 0.00144913.$$

When the filter length increased from 29 to 39, both the passband ripple and the stopband ripple decreased. I also found that the passband edge ω_p moved closer to the desired cutoff frequency 0.33π, and the stopband edge ω_s moved inward toward the cutoff as well. Hence increasing the filter length improved the approximation of the ideal lowpass response by reducing ripple and sharpening the transition behavior.

Code

```
%% Exercise 12: Optimal FIR vs. Elliptic IIR Design
% Replacement: FIR frequency-sampling via fir2 (N=29 and N=39)

clear; close all; clc;

% Specs
wc   = 0.33;
f    = [0 wc wc 1];
m    = [1 1 0 0];
Nfft = 8192;

Ns = [29 39];

for k = 1:numel(Ns)
    N = Ns(k);
    ord = N-1;

    %% Design (fir2 frequency-sampling FIR)
    h = fir2(ord, f, m);
```

```
    %% Frequency response
    [H,w] = freqz(h, 1, Nfft);
    mag = abs(H);
    wn  = w/pi;

    %% Impulse response
    n = 0:ord;
    figure('Color','w','Name',sprintf('Impulse Response (fir2, N=%d)',N));
    stem(n, h, 'filled'); grid on;
    xlabel('n'); ylabel('h[n]');
    title(sprintf('fir2 Lowpass, N=%d, \\omega_c=%.2f\\pi', N, wc));

    %% Magnitude response (linear)
    figure('Color','w','Name',sprintf('Magnitude Response (fir2, N=%d)',N));
    plot(wn, mag); grid on; xlim([0 1]);
    xlabel('\omega/\pi'); ylabel('|H(e^{j\omega})|');
    title(sprintf('fir2 Magnitude (Linear), N=%d', N));

    % I need to use zoom on this plot to read off:
    %   delta_p1, delta_p2, omega_p, delta_s, omega_s
    % following the same definitions as Exercises 5/6.
end
```